

ENTITY RELATIONSHIP DIAGRAM (ERD)

Company Internal Reporting System

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Document	ERD
Database	PostgreSQL

1. Introduction

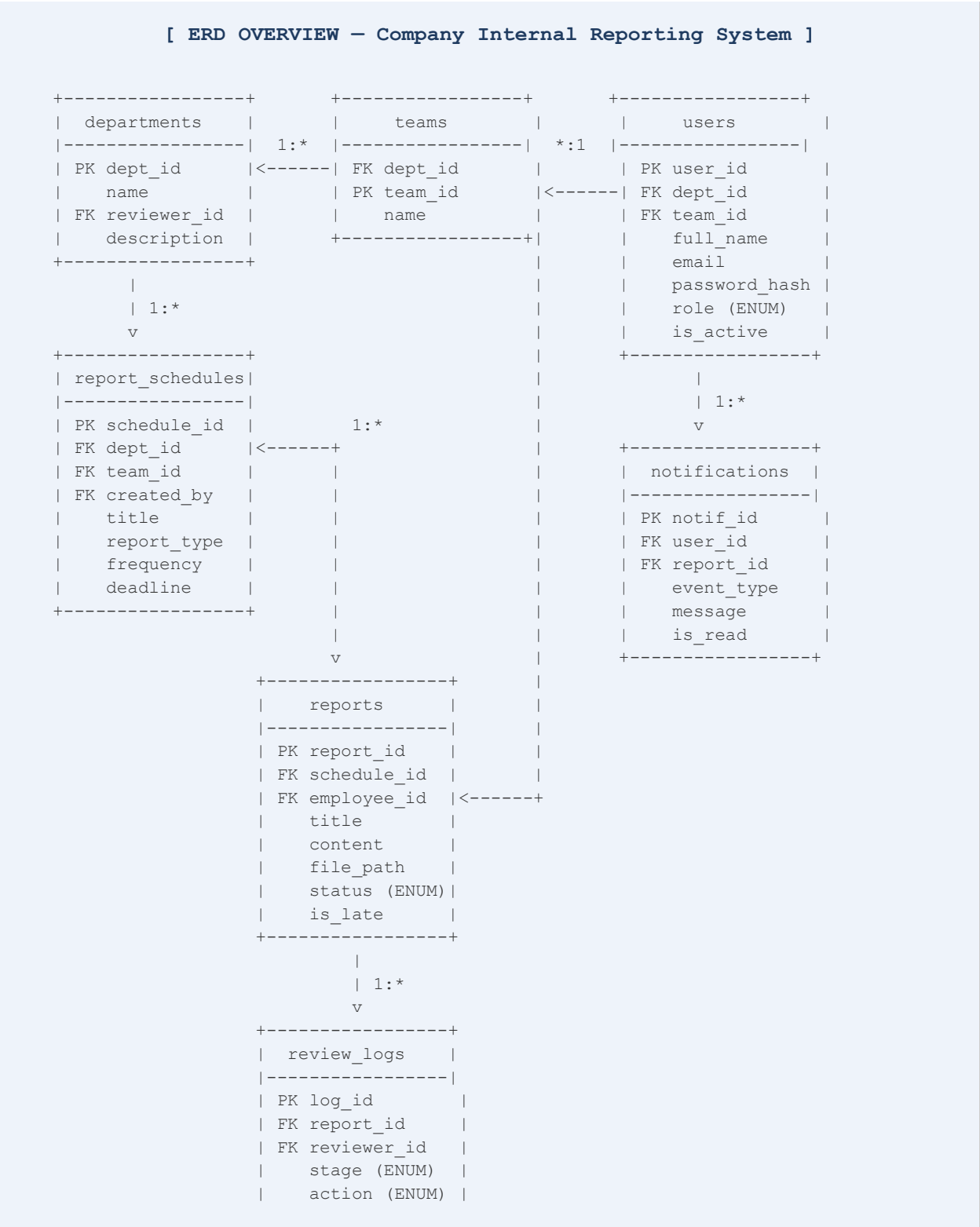
This document defines the Entity Relationship Diagram (ERD) for the Company Internal Reporting System. It describes all database entities, their attributes, primary and foreign keys, and the relationships between them. This serves as the direct blueprint for the PostgreSQL database schema.

1.1 Notation Key

Symbol / Term	Meaning
PK	Primary Key — uniquely identifies each record in a table
FK	Foreign Key — references the primary key of another table
NOT NULL	The column must always have a value
UNIQUE	All values in the column must be distinct
DEFAULT	A default value is applied if none is provided
SERIAL	Auto-incrementing integer, managed by PostgreSQL
ENUM	Column restricted to a predefined set of string values
1 : Many	One record in table A relates to many records in table B
1 : 1	One record in table A relates to exactly one in table B

2. ERD Overview Diagram

The diagram below shows all 7 entities and their relationships. Arrows show foreign key references.



review_logs

PK log_id

FK report_id

FK reviewer_id

stage (ENUM)

action (ENUM)

	comment	
	created_at	
+-----+		

3. Entity Schemas

Each entity maps directly to a PostgreSQL table.

3.1 departments

Stores all company departments. Each department has one assigned reviewer (a user with role='reviewer').

departments			
Column Name	Data Type	Key	Notes
dept_id	SERIAL	PK	Auto-incremented, unique department ID
name	VARCHAR(100)		Unique department name, NOT NULL
description	TEXT		Optional description
reviewer_id	INTEGER	FK	References users(user_id) — dept reviewer
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		Auto-updated on change

3.2 teams

Stores teams belonging to departments. Employees are assigned to teams.

teams			
Column Name	Data Type	Key	Notes
team_id	SERIAL	PK	Auto-incremented unique ID
name	VARCHAR(100)		Unique within department, NOT NULL
dept_id	INTEGER	FK	References departments(dept_id), NOT NULL
created_at	TIMESTAMP		DEFAULT NOW()

3.3 users

Stores all system users: Admin, Employee, Department Reviewer, and Final Approver.

users			
Column Name	Data Type	Key	Notes
user_id	SERIAL	PK	Auto-incremented unique ID
full_name	VARCHAR(150)		NOT NULL
email	VARCHAR(255)		UNIQUE, NOT NULL — used for login and notifications
password_hash	VARCHAR(255)		bcrypt-hashed password, NOT NULL
role	user_role (ENUM)		admin employee reviewer approver — NOT NULL
dept_id	INTEGER	FK	References departments(dept_id), nullable for admin/approver
team_id	INTEGER	FK	References teams(team_id), nullable
is_active	BOOLEAN		DEFAULT TRUE — soft deactivation
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		Updated on profile change

3.4 report_schedules

Admin-defined schedules that determine what reports employees must submit and when.

report_schedules			
Column Name	Data Type	Key	Notes
schedule_id	SERIAL	PK	Auto-incremented unique ID
title	VARCHAR(200)		Descriptive schedule name, NOT NULL
report_type	VARCHAR(100)		E.g. Monthly Financial, Weekly Progress
frequency	frequency_type (ENUM)		weekly biweekly monthly quarterly custom
start_date	DATE		When the schedule period begins, NOT NULL

deadline	DATE		Report submission deadline, NOT NULL
dept_id	INTEGER	FK	References departments(dept_id), nullable
team_id	INTEGER	FK	References teams(team_id), nullable
created_by	INTEGER	FK	References users(user_id) — admin who created
created_at	TIMESTAMP		DEFAULT NOW()

3.5 reports

Central table storing all employee report submissions with their full lifecycle status.

reports			
Column Name	Data Type	Key	Notes
report_id	SERIAL	PK	Auto-incremented unique ID
schedule_id	INTEGER	FK	References report_schedules(schedule_id), NOT NULL
employee_id	INTEGER	FK	References users(user_id) — submitting employee, NOT NULL
title	VARCHAR (255)		Report title, NOT NULL
content	TEXT		Inline report body; nullable if file uploaded
file_path	VARCHAR (500)		Path or URL to uploaded file; nullable
file_name	VARCHAR (255)		Original uploaded filename
status	report_status (ENUM)		pending submitted under_review changes_requested approved rejected
is_late	BOOLEAN		DEFAULT FALSE — TRUE if submitted after deadline
submitted_at	TIMESTAMP		Actual timestamp of submission
created_at	TIMESTAMP		DEFAULT NOW()
updated_at	TIMESTAMP		Updated on any status change

3.6 review_logs

Audit trail for every review action on a report, covering both Stage 1 and Stage 2.

review_logs			
Column Name	Data Type	Key	Notes
log_id	SERIAL	PK	Auto-incremented unique ID
report_id	INTEGER	FK	References reports(report_id), NOT NULL
reviewer_id	INTEGER	FK	References users(user_id) — reviewer/approver, NOT NULL
stage	review_stage (ENUM)		stage_1 stage_2 — NOT NULL
action	review_action (ENUM)		approved rejected changes_requested — NOT NULL
comment	TEXT		Optional for approvals; REQUIRED for rejections
created_at	TIMESTAMP		DEFAULT NOW() — immutable audit timestamp

3.7 notifications

In-app notification records for all system users, triggered by workflow events.

notifications			
Column Name	Data Type	Key	Notes
notif_id	SERIAL	PK	Auto-incremented unique ID
user_id	INTEGER	FK	References users(user_id) — recipient, NOT NULL
report_id	INTEGER	FK	References reports(report_id), nullable
event_type	VARCHAR(50)		E.g. report_due, submitted, approved, rejected
message	TEXT		Notification body text, NOT NULL

is_read	BOOLEAN		DEFAULT FALSE — marks as read when user opens it
created_at	TIMESTAMP		DEFAULT NOW()

4. Relationships Summary

Relationship	Type	FK Column	References
departments → teams	1 : Many	teams.dept_id	departments(dept_id)
departments → users (reviewer)	1 : 1	departments.reviewer_id	users(user_id)
departments → users (employees)	1 : Many	users.dept_id	departments(dept_id)
teams → users	1 : Many	users.team_id	teams(team_id)
departments → report_schedules	1 : Many	report_schedules.dept_id	departments(dept_id)
teams → report_schedules	1 : Many	report_schedules.team_id	teams(team_id)
users → report_schedules (created_by)	1 : Many	report_schedules.created_by	users(user_id)
report_schedules → reports	1 : Many	reports.schedule_id	report_schedules(schedule_id)
users (employee) → reports	1 : Many	reports.employee_id	users(user_id)
reports → review_logs	1 : Many	review_logs.report_id	reports(report_id)
users (reviewer) → review_logs	1 : Many	review_logs.reviewer_id	users(user_id)
users → notifications	1 : Many	notifications.user_id	users(user_id)
reports → notifications	1 : Many	notifications.report_id	reports(report_id)

5. ENUM Type Definitions

Custom PostgreSQL ENUM types used in the schema:

ENUM Type	Values	Used In Column
user_role	admin, employee, reviewer, approver	users.role
report_status	pending, submitted, under_review, changes_requested, approved, rejected	reports.status
frequency_type	weekly, biweekly, monthly, quarterly, custom	report_schedules.frequency
review_action	approved, rejected, changes_requested	review_logs.action
review_stage	stage_1, stage_2	review_logs.stage

6. Recommended Database Indexes

Index Name	Table	Column(s)	Purpose
idx_users_email	users	email	Fast authentication lookup
idx_users_dept	users	dept_id	Filter employees by department
idx_reports_employee	reports	employee_id	Get all reports by a user
idx_reports_status	reports	status	Dashboard filtering by status
idx_reports_schedule	reports	schedule_id	Join reports to their schedule
idx_review_logs_report	review_logs	report_id	Retrieve full audit history for a report
idx_notifications_user	notifications	user_id, is_read	Fetch unread notifications per user
idx_schedules_dept	report_schedules	dept_id	Get all schedules for a department